INCH-POUND

MIL-DTL-83513/10D
29 October 2008
SUPERSEDING
MIL-DTL-83513/10C
20 September 2002

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, PIN CONTACTS, 2 ROW, SOLDER TYPE, NARROW PROFILE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83513.

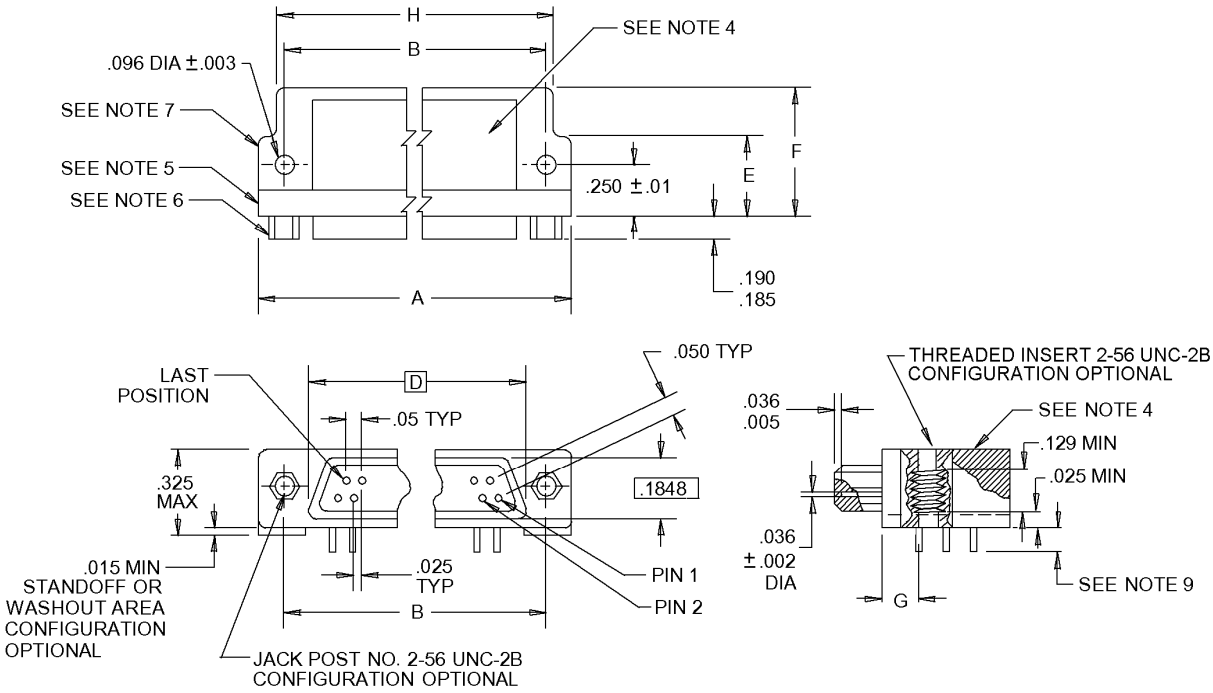
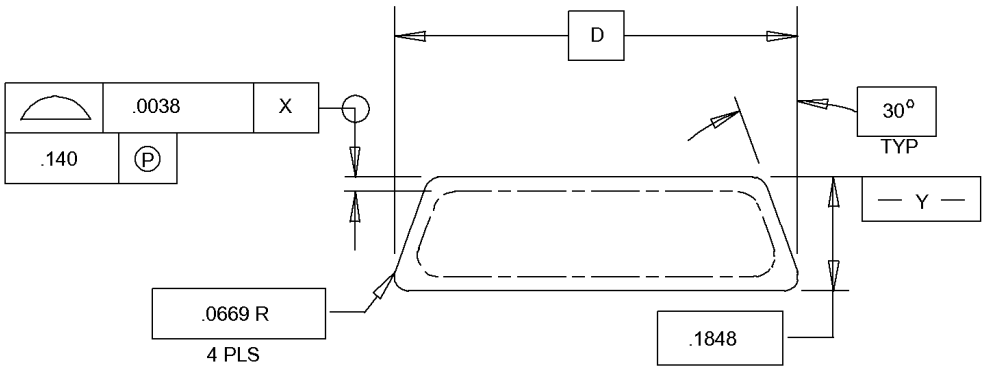


FIGURE 1. Connector, plug, .050 spacing.

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DETAIL OF INTERFACE

Inches	mm	Inches	mm
.002	0.05	.109	2.77
.003	0.08	.140	3.56
.0038	0.097	.1848	4.694
.010	0.25	.185	4.70
.015	0.38	.186	4.72
.036	0.91	.190	4.83
.050	1.27	.250	6.35
.0669	1.699	.325	8.26
.096	2.44		

FIGURE 1. Connector, plug, .050 spacing - Continued.

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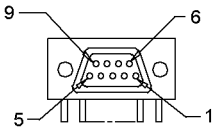
Number of contacts	A Max	B ±.005	D	E Max	F Max	G ±.01	H Max
9	.787 (19.99)	.565 (14.35)	.3338 (8.48)	.425 (10.80)	.425 (10.80)	.230 (5.84)	.787 (19.99)
15	.937 (23.80)	.715 (18.16)	.4838 (12.29)	.425 (10.80)	.425 (10.80)	.130 (3.30)	.937 (23.80)
21	1.087 (27.61)	.865 (21.97)	.6338 (16.10)	.425 (10.80)	.425 (10.80)	.130 (3.30)	1.087 (27.61)
25	1.187 (30.15)	.965 (24.51)	.7338 (18.64)	.425 (10.80)	.425 (10.80)	.130 (3.30)	1.187 (30.15)
31	1.337 (33.96)	1.115 (28.32)	.8838 (22.45)	.450 (11.43)	.525 (13.34)	.130 (3.30)	1.090 (27.69)
37	1.487 (37.77)	1.265 (32.13)	1.0338 (26.26)	.450 (11.43)	.525 (13.34)	.130 (3.30)	1.190 (30.23)

NOTES:

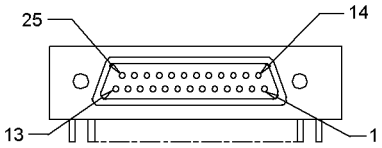
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-DTL-83513. Plastic molding shall conform to type GDI-30F or type SDG-F in accordance with ASTM D5948 or GCT-30F in accordance with ASTM D5927 or MIL-M-24519 or GST-40F in accordance with ASTM D4067 or MIL-M-24519 or GLCP-30F or GLCP-50 in accordance with ASTM D5138 or MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-DTL-83513 for class M parts.
6. Jackpost (securely attached) when specified: Corrosion resistant steel in accordance with ASTM A484/A484M and ASTM A582/A582M, 300 series stainless steel, passivated in accordance with SAE-AMS-2700, type 2. Hardware will be attached in such a fashion as to not loosen during normal de-mating conditions, where as, the mating connector hardware does not exceed the following torque values: 2.5 inch pounds for #2-56 (arrangements A – G) and 4.5 inch pounds for #4-40 (arrangement H).
7. Separately molded plastic body (if used) shall conform to type GDI-30F or type SDG-F in accordance with ASTM D5948 or GCT-30F in accordance with ASTM D5927 or MIL-M-24519 or GST-40F in accordance with ASTM D4067 or MIL-M-24519 or GLCP-30F or GLCP-50 in accordance with ASTM D5138 or MIL-M-24519.
8. Wire termination pins shall conform to A-A-59551, number 24 AWG copper, except shall be tin-lead plated with a minimum of 3 percent lead.
9. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm) or .172 (4.37 mm). The tolerance shall be ± .015 (0.381 mm) for all termination lengths.
10. Threaded insert, when specified: Corrosion resistant steel in accordance with ASTM A484/A484M and ASTM A582/A582M, 300 series stainless steel, passivated in accordance with SAE-AMS-2700, type 2.

FIGURE 1. Connector, plug, .050 spacing. – Continued.

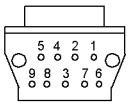
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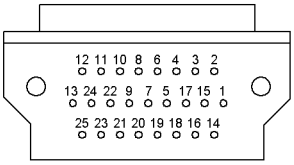
9 CAVITY



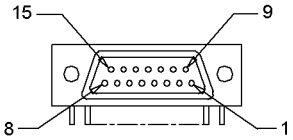
25 CAVITY



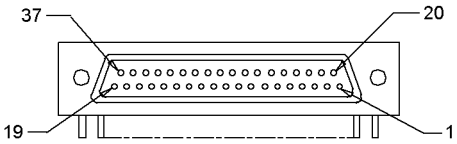
TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE



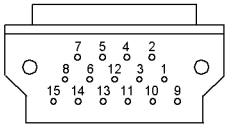
TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE



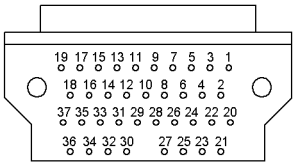
15 CAVITY



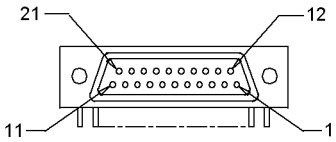
37 CAVITY



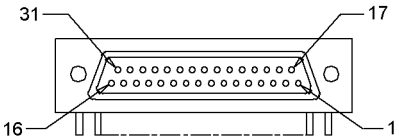
TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE



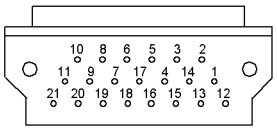
TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE



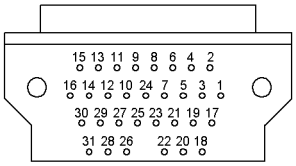
21 CAVITY



31 CAVITY



TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE



TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE

NOTE: Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement, termination view - right angle - narrow profile.

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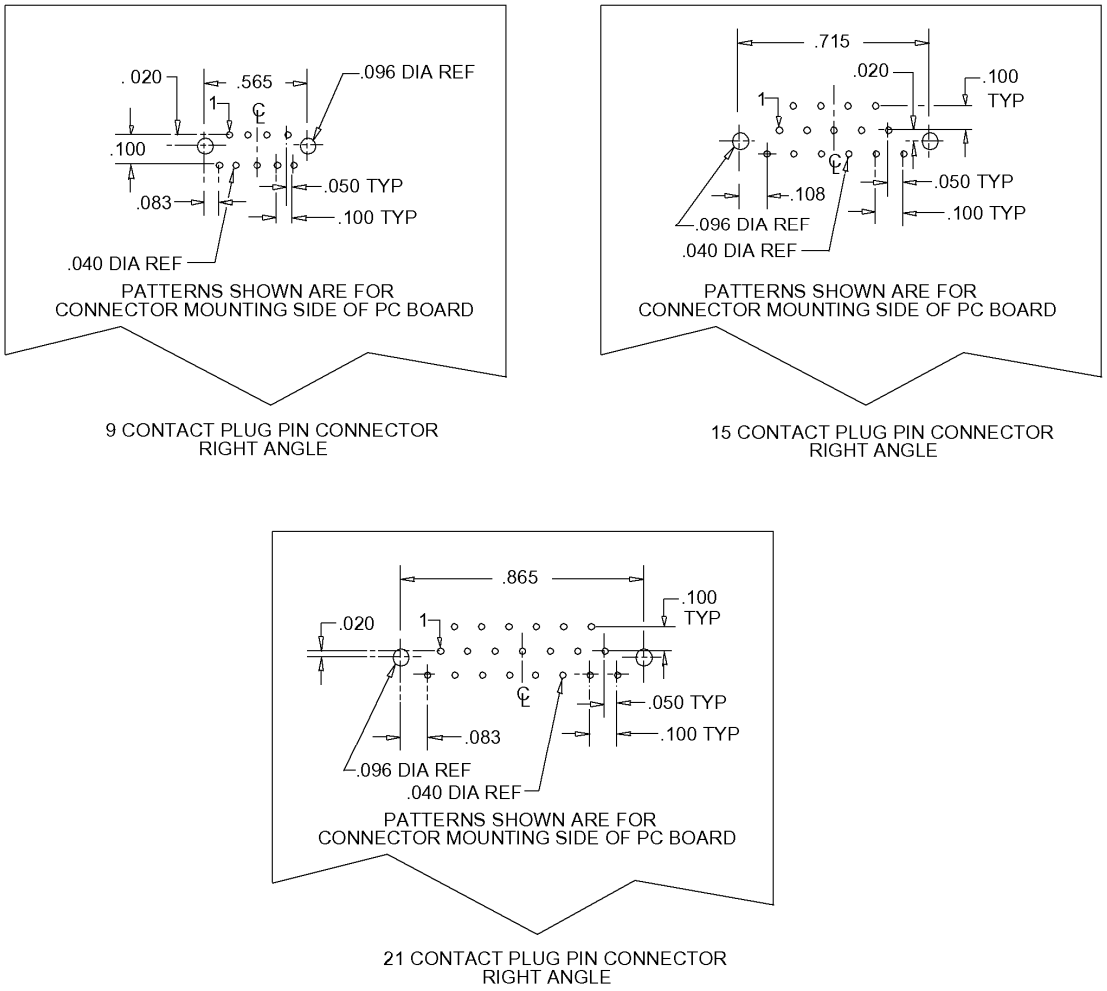
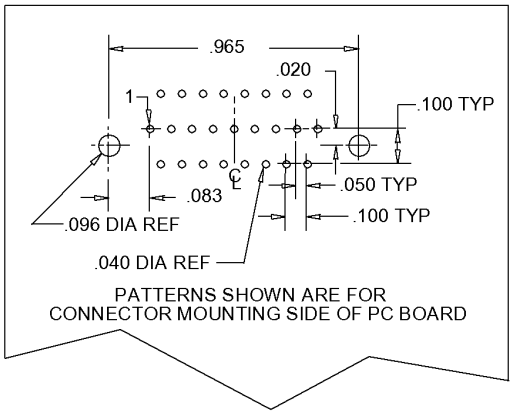
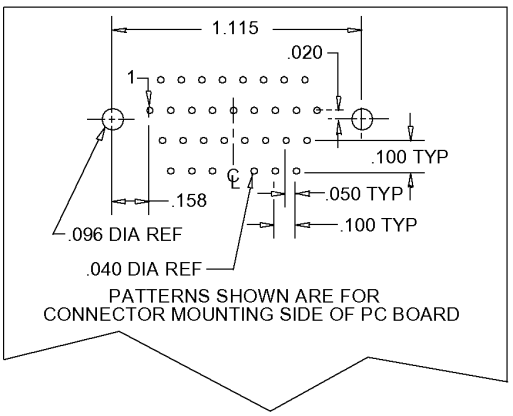


FIGURE 3. Layout arrangement.

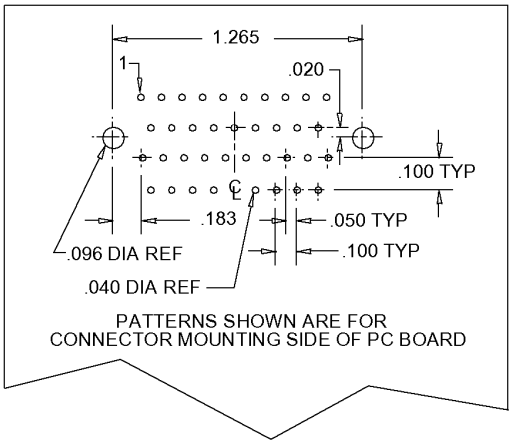
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25 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE



31 CONTACT PLUG PIN CONECTOR
RIGHT ANGLE



37 CONTACT PLUG PIN CONNECTOR
RIGHT ANGLE

Inches	mm	Inches	mm
.020	0.51	.158	4.01
.040	1.02	.565	14.35
.050	1.27	.715	18.16
.083	2.11	.865	21.97
.096	2.44	.965	24.51
.100	2.54	1.115	28.32
.108	2.74	1.265	32.13

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm).

FIGURE 3. Layout arrangement - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figures 1, 2 and 3.

Current rating, maximum: 3 amperes per contact.

Materials:

Termination organization area: Potting fill material capable of passing the electrical and environmental requirements of MIL-DTL-83513.

Shell: The requirements for shell materials shall be in accordance with MIL-DTL-83513.

Plastic body or plastic molding: Shall conform to the requirements of GDI-30F or type SDG-F in accordance with ASTM D5948 or GCT-30F in accordance with ASTM D5927 or MIL-M-24519 or GST-40F in accordance with ASTM D4067 or MIL-M-24519 or GLCP-30F or GLCP-50 in accordance with ASTM D5138 or MIL-M-24519.

Jackpost: Corrosion resistant steel in accordance with ASTM A484/A484M and ASTM A582/A582M, 300 series stainless steel, passivated in accordance with SAE-AMS-2700, type2.

Wire termination pins: Wire termination pins shall conform to A-A-59551, no. 24 AWG copper, except shall be tin-lead plated with a minimum of 3 percent lead.

Mating connector: Shall conform to MIL-DTL-83513/2 and MIL-DTL-83513/4.

Plating of termination leads: Solder dipping of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with J-STD-006.

Part or Identifying Number (PIN): PIN shall consist of the letter M, the basic number of the specification sheet, a letter from the insert, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/10-	D	01	A	P
Specification sheet number	Insert arrangements (see figure 2)	Termination length	Shell finish (interface critical)	Hardware
	A = 09 B = 15 C = 21 D = 25 E = 31 F = 37	01 = .109 02 = .140 03 = .172	A = Pure electrodeposited aluminum C = Cadmium K = Zinc nickel N = electroless nickel (space applications only) P = Passivated Stainless Steel T = Nickel fluorocarbon polymer	N = No hardware or threaded insert P = jackpost attach T = threaded insert W = jackpost and threaded insert

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Reference documents. In addition to MIL-DTL-83513, this document references the following:

MIL-DTL-83513/2	ASTM A484/A484M	A-A-59551
MIL-DTL-83513/4	ASTM A582/A582M	J-STD-006
MIL-M-24519	ASTM D4067	SAE-AMS-2700
	ASTM D5138	
	ASTM D5927	
	ASTM D5948	

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AT, MI
Navy - AS, CG, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-2007-058)

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